

**Questionnaire for Member States' reporting on the implementation of the
UNESCO 2021 Recommendation on Open Science**

**1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,
ASSOCIATED WITH BENEFITS AND CHALLENGES AS WELL AS DIVERSE
PATHS TO OPEN SCIENCE**

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

No

Although the relevant government agencies, ministries, and their research and academic institutions have access to the proposed Recommendations on the concept of Open Science, the mentioned Recommendation document has not been explicitly shared with state institutions and scientific organizations.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

No

1.3 Have there been awareness raising activities on open science, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref: (i) 16 f, 16 g, 16 h)

Yes.

In Azerbaijan, numerous awareness-raising activities have been held on open science in accordance with the 16f, 16g, and 16h paragraphs of the Recommendation. While policy administrators and scientists are informed about open science collaboration, the general public has limited awareness about it. The private sector also lacks sufficient knowledge on open science collaboration; however, it has a certain share in the funding of science in Azerbaijan. Taking the mentioned points into account, Higher Education Institutions (HEIs) and research institutes under the Ministry of Science and Education of the Republic of Azerbaijan (MoSE), have called on the scientific community to share their research findings with the public. In 2025, relevant bodies of the Ministry of Science and Education

of the Republic of Azerbaijan aim to promote the importance of the principles and benefits of open science.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the value and principles of open science, in publicly funded research? (ref: (i) 16a, 16b, 16c, 16d, 16e, 16h)

Yes.

Relevant activities have been and are being carried out within the framework of a "Roadmap for Improvement of the international rating indicators related to scientific publications (H-index)," under the leadership of the Cabinet of Ministers of the Republic of Azerbaijan:

- On January 15, 2020, the Institute of Education (IoE) hosted an Open Science Workshop conducted by international experts. The workshop brought together 60 participants from different public and private universities, including regional ones, the Institute of Education and Intellectual Property Agency, targeting mainly science departments, sectors dealing with scientific publications in HEIs, and library experts.
- On May 5, 2023, the Institute of Physics under the Ministry of Science and Education of the Republic of Azerbaijan held a seminar on "Open Science." Participants were provided with detailed information on the ground principles of open science, as well as existing achievements and challenges in this field.
- On October 31, 2024, the Azerbaijan Society of Young Scientists, Postgraduates, and Masters, with the support of the Ministry of Youth and Sports, organized a meeting in Lankaran city, the "Youth Capital" of 2024, as part of the project "Open Science Summer School for Young Researchers." During the event, young researchers were informed about access to open scientific resources, methods for effectively using research databases, methods for preparing scientific articles, and open educational resources through various presentations.

HEIs are also actively working in this direction:

- On 9 February 2022, Baku State University (BSU) with the support of the Ministry of Education of the Republic of Azerbaijan and Azerbaijan Young Scientists, Post-Graduates and Masters Union (AYSPMU) hosted the workshop "Open Science

Azerbaijan: Promotion of the Open Science platform among young researchers” within the framework of the Open Science Azerbaijan project. The main goal of the project was to inform young researchers about Open Science, Open Access, Open Research Data, Open Innovation, Open Research Computation, ResearchGate, Scientific Social Network.

- On February 17, 2022, Azerbaijan University of Languages hosted a seminar on "Open Science Azerbaijan: Promoting the Open Science Platform Among Young Researchers" with joint support from the "Open Science Azerbaijan" project.
- On February 28, 2022, Azerbaijan University of Technology, the Ministry of Education and the Azerbaijan Society of Young Scientists, Postgraduates, and Masters jointly organized a seminar, as part of the project "Open Science Azerbaijan: Promoting the Open Science Platform Among Young Researchers."
- On December 6, 2024, the event titled "Open Access and Science Diplomacy" was held within the framework of the "Erasmus+ KA2" program's "Development of Research Potential through the Institutional Repositories Network in Azerbaijan" (DIRNA) project at Azerbaijan Technical University.
- On December 23, 2024, the International Open Science and Education Laboratory was opened under the Pedagogy Department at Baku State University (BSU). The resources of the laboratory will be used both in the educational process and for the correction of students with special educational needs, focusing on their integration into society.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy, strategy or plan of action on science, technology and innovations (STI)?

Yes.

The elements of open science are reflected in the relevant documents, including the Law of the Republic of Azerbaijan "On Science,"¹ adopted in 2016. Additionally, the "Roadmap for Improvement of the international rating indicators related to scientific publications (H-index)"² has been developed. The roadmap has five

¹ 271-VQ - Elm haqqında

1009 - "Elm haqqında" Azərbaycan Respublikasının 2016-cı il 14 iyun tarixli 271-VQ nömrəli Qanununun tətbiqi barədə

² H-index görülmüş işlər_V5.3.pdf

objectives, such as integration into the global scientific space, strengthening scientific intellectual potential, formation of incentive mechanisms, defining clear targets and evaluation policies for state scientific research and aligning scientific publications with international standards

Under these five objectives, a total of 35 activities have been carried out in 17 areas in 2023. As part of the necessary measures taken for the implementation of relevant proposals on the centralized funding of access to international scientific databases for science and higher education institutions, more than 100 science and education institutions have gained access to the Scopus and Science Direct databases, as well as the Web of Science scientometric database, and have been informed accordingly. In 2024, the number of publications by Azerbaijani scientists in journals indexed in the Scopus scientometric database, owned by Elsevier, increased by nearly 40% compared to the previous year and amounted to 4.088 publications. Additionally, with the support of the State Agency for Science and Education (SASHE), Elsevier and Clarivate carried out this first-ever joint training program in scientometrics. The program aimed to train specialists capable of applying Scopus and Web of Science scientometric databases at an expert level in the country.

2.2 In your country, is there a policy or a set of policies and/or legal framework(s) at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref (i) 16a, 16b, 16c, 16d, 16e, 16h)

On a legal basis, the Law of the Republic of Azerbaijan "On Science," adopted in 2016, defines the principles, goals, and priorities of state policy in the field of scientific activity. According to this document, the values and principles of science in Azerbaijan align with those outlined in the 2021 UNESCO Recommendation on Open Science. State-funded scientific research is open to the public and is primarily financed by the Science Fund. According to Article 6.0.32 of the Law "On Science," it is the duty of the state of Azerbaijan to prevent the development and application of scientific technologies that contradict humanism, endanger human life, disrupt societal stability, threaten national security, or harm ecological balance. This fully complies with the directions stated in Article 8 of the UNESCO

Recommendation on Open Science which proposes open access to scientific knowledge, with access restrictions allowed only for legitimate reasons such as protecting human rights, privacy, intellectual property, and sensitive data.

The official language of the Republic of Azerbaijan is Azerbaijani language. However, multilingualism in the practice of science, in scientific publications, and in academic communications is encouraged, as scientific research in the country is conducted in Azerbaijani, English, and Russian. It is a mandatory requirement that abstracts for obtaining scientific degrees be prepared in three languages (Azerbaijani, English, and Russian). Azerbaijan also provides the Recommendation on prevention of violation of the needs and rights of communities, including the rights of indigenous peoples over their traditional knowledge in open science practices.

2.3 In your country, are there specific policy instruments that aim to promote open science?

Yes.

The Law of the Republic of Azerbaijan "On Science" is referred to as a policy tool.

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref (ii) 17a, 17b, 17c, 17f, 17h)

Yes.

Although the term "Open Science" is not explicitly used in the existing legislative framework, the relevant documents such as the Law "On Science", "Roadmap for Improvement of the international rating indicators related to scientific publications (H-index)" are in line with paragraph 17 the Recommendation. This roadmap entails five objectives including, integration into the global scientific space, strengthening scientific intellectual potential, formation of incentive mechanisms, defining clear targets and evaluation policies for state scientific research and aligning scientific publications with international standards. Azerbaijan regularly carries out numerous activities within the framework of these directions.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref (ii) 17d, 17e, 17g)

Yes

Although the term "Open Science" is not explicitly used in the existing legislative framework, the relevant documents such as the Law "On Science", " Roadmap for Improvement of the international rating indicators related to scientific publications (H-index)" are in line with paragraph 17 the Recommendation.

Azerbaijan has implemented various initiatives to promote open science at the institutional level. Research-performing institutions, universities, and scientific associations support open science through various workshops, seminars, as well as capacity-building projects. Specific efforts include the "Open Science Azerbaijan" project, which held workshops in 2022 at multiple universities to educate young researchers on open access, open data, and scientific networks. In 2024, the "Open Access and Science Diplomacy" event at Azerbaijan Technical University and the establishment of the International Open Science and Education Laboratory at Baku State University further emphasized institutional engagement.

2.6 In your country, are there specific funding mechanisms for open science at the national level? (ref: (ii), 17 j, (iii) 18 j, (iv) 19 c)

No

2.7 In your country, have open science practices been integrated in existing research funding mechanisms?

Yes

According to Article 34 of the Law of the Republic of Azerbaijan "On Science", the financing of scientific and scientific-technical activities from the state budget is carried out in the following forms:

- Base financing
- Program financing
- Grant financing

The foundations of program-targeted financing are outlined in Article 36 of the Law "On Science." According to the Law, program-targeted financing for scientific activities is directed towards the implementation of state-significant scientific work of strategic importance and is carried out either through competition or, based on the decision of the relevant executive authority, without competition. Regardless of ownership type, all subjects of scientific activity and their relevant structures have equal rights when participating in the competition for program-targeted financing.

The subjects of base financing for scientific activities are state scientific institutions and organizations, scientific research institutes and laboratories of state higher education institutions, as well as other scientific institutions fulfilling state orders in scientific activities. Base financing ensures the conduct of fundamental scientific research, state-important research, including those related to national security and defense, the preservation of important scientific objects, and the training of highly qualified scientific personnel.

Grant financing is carried out to stimulate scientific research, increase its efficiency, raise the level of scientific-technical potential, and enhance the competitiveness of scientific institutions and organizations, higher education institutions, as well as scientists and researchers.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnership on open science in line with the 2021 Recommendation? (ref: (i) 17 i)

Yes

In the Republic of Azerbaijan, public-private collaboration is continuously carried out in the fields of science and education. Although this collaboration is more prevalent in the education sector, in recent years, institutional-level collaboration has also begun. Various private companies are participating in the financing of scientific research:

- BP Azerbaijan has implemented an educational project aimed at strengthening the scientific research potential of the Azerbaijan State University of Economics (UNEC). Within the framework of this project, research equipment, research trips

for UNEC's scientists, and the expenses of three joint conferences were financed by the company.

- Companies such as "Elroys" LLC and "IselTexno" LLC are closely collaborating with the Institute of Chemistry of Additives after Academician A.M.Guliyev in this field. A significant result of the collaboration with "Elroys" LLC is the creation and application of the "Depressator AKI" additive. This additive is one of the most successful applied works of the Institute. It ensures the transportation of high-paraffin oil from Kazakhstan and Turkmenistan through the Baku-Tbilisi-Ceyhan pipeline over long distances by lowering their freezing point and dynamic viscosity. With the financial support of "Elroys" LLC, its production is now organized at the "Additive" Scientific Production Union, and it is being successfully used.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy? (ref (ii) 17j, 23c)

No

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref (iii) 18a)

According to the "2022–2026 Social and Economic Development Strategy" of the Republic of Azerbaijan, the share of science expenditure in the Gross Domestic Product (GDP) is 0.2%. The goal is to reach the global average share of GDP spent on scientific research and development projects by 2026³.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data and to which year(s) it refers. (ref (iii) 18b)

No.

Statistical analysis has not been carried out in this regard.

³ E-QANUN

3.3 Are there national research and education networks (NRENs) active in the country?(ref (iii) 18c)

No

The work related to the digitalization of science and education in the country started only in 2023. Hence, no relevant system exists at the moment.

3.4 Are there national or institutional open science infrastructures in your country? (ref (iii) 18d, 18e, 18k, 18l, 18m)

Under development

Certain efforts have been made in this regard. Technoparks with the necessary infrastructure and material-technical base for conducting scientific research and experimental design work, as well as for applying their results in industry, services, and other fields, have been created. These areas are open to scientists, and information about the research infrastructure owned by the state or created with state funds has been summarized, providing researchers with information about the availability of relevant equipments.⁴

3.5 Does your country host or fund and federated regional or international information technology infrastructure for open science? (ref (iii) 18e, 18f)

No

At the same time, efforts are being made to provide Azerbaijani scientists with access to international systems. For example, in 2015, an agreement ⁵was signed between the Ministry of Education of the Republic of Azerbaijan and the Scientific Research and Intellectual Property Department of "Thomson Reuters." On June 14, 2023, a contract was signed between the Agency for Quality Assurance in Education (AQAE) and "Elsevier" to provide access to the world's largest bibliometric database, "Scopus," for private and state higher education institutions as well as all research institutes in Azerbaijan. With the signing of this contract, more than 100 state and private institutions, researchers, and students in

⁴ 449 - "Dövlət mülkiyyətində olan və ya dövlət vəsaiti hesabına yaradılan tədqiqat infrastrukturunun əvədlilikindən istifadəçilərin istifadəsinə görə ödənilən xidmət haqları"nın təsdiq edilməsi barədə

⁵ anl.az/down/meqale/az_muellimi/2015/dekabr/470299.htm

Azerbaijan now have access to various resources and tools in the "Scopus" database.

3.6 Is your country involved in any North-South, North-South-South, and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms?(ref (iii) 18g)

No

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country?(ref (iv) 19a, 19c)

No

Currently, there is no systematic work being held on open science, but Azerbaijani science is open to the subjects of science at all scientific and higher education institutions.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

Yes

In 2015, a cooperation agreement was signed for the first time between the Ministry of Education of the Republic of Azerbaijan and the Scientific Research and Intellectual Property Department of "Thomson Reuters" company. Relevant training sessions are being conducted in this regard. Organization of such training sessions for individuals who shape the policy is under consideration.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your-country, or is it foreseen by the end of 2025? (ref (iv) 19b)

No

Although topics related to "Open Science" are not included in the curricula, it is planned to develop educational programs for the master's level this year, and

relevant competencies related to this topic are expected to be incorporated into those programs.

4.4 Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref (iv) 19d)

Partly

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref (iv) 19e)

Yes

The Ministry of Science and Education of the Republic of Azerbaijan continuously supports the establishment and development of communications in the scientific field. For example, in order to organize discussions on the results of scientific research conducted by doctoral students and young researchers, as well as to stimulate their scientific creativity, the Ministry holds the annual Republican Scientific Conference for Doctoral Students and Young Researchers. Additionally, the state has established the "Presidential Award for Youth" ("Gənclər üçün Prezident Mükafatı")⁶ and the "Youth Award" ("Gənclər mükafatı") to encourage and recognize young talents.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end 2025?(ref (v) 20b, 20c, 20i)

Yes

An accreditation procedure is carried out in order to determine the quality and effectiveness of scientific programs and the results of scientific research. This

⁶ 934 - Gənclər üçün Prezident Mükafatının təsis edilməsi haqqında

procedure is conducted based on the rules approved by the Cabinet of Ministers of the Republic of Azerbaijan under the decision⁷ No. 554 dated December 14, 2017.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref (v) 20a, 20c, 20d)

Yes

Student Scientific Societies⁸ have been established in higher education institutions to engage young people in the field of science. The Student Scientific Society is a voluntary public organization of students who actively participate in scientific research and scientific-organizational activities at the bachelor's and master's levels of higher education, as well as at medical education institutions and in residency programs in the field of medical education.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref (v) 20a, 20b, 20c, 20e, 20f, 20g, 20h)

Yes

Azerbaijani government supports initiatives aimed at increasing the number of articles published by Azerbaijani scientists in international journals with impact factors. Within this framework, as a result of cooperation between HEIs, research institutes, and the industry, the total number of articles published in international journals with impact factors (Scopus) has increased, particularly the number of articles published in Q1 and Q2 ranked journals. To achieve this goal, a financial incentive mechanism for researchers has been implemented.

5.4 Have there been any unintended negative consequences of open science practices reported in your country? (ref (v) 20)

No

⁷ 554 - "Elmi müəssisələrin təşkilatların akkreditasiyası Qaydası"nın təsdiq edilməsi haqqında

⁸ Tələbə Elmi Cəmiyyəti

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process?

No

By the Decree of the President of the Republic of Azerbaijan dated July 28, 2022, the authority to implement state policy and regulation in the field of science, as well as to coordinate and direct the scientific activities of scientific institutions and organizations, has been entrusted to the Ministry of Science and Education of the Republic of Azerbaijan. It is hence planned to prepare a document of an action plan in the current year.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science?

As mentioned in previous questions, international cooperation in the field of open science is primarily carried out at the institutional level, with various scientific and higher education institutions in our country implementing projects in this area.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science including with regards to exchange of best practices, capacity building and metrics for open science?

No.

By the Decree of the President of the Republic of Azerbaijan dated July 28, 2022, the authority to implement state policy and regulation in the field of science, as well as to coordinate and direct the scientific activities of scientific institutions and organizations, has been entrusted to the Ministry of Science and Education of the Republic of Azerbaijan. It is hence planned to prepare a document of an action plan in the current year.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?

No

It is planned to conduct relevant research in this area in the future.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

No